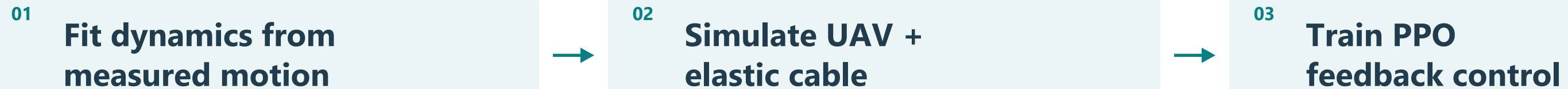


UAV cable-tip targeting

Move the UAV so the cable tip hits the target with less UAV travel.

Control: horizontal acceleration, vertical acceleration and pitch rate, updated at 10 Hz.

HOW IT WORKS



CABLE MODEL

12-node elastic rod; 10 free nodes. Identified bending and damping. Cable-to-UAV reaction is currently omitted.

FULL STATE?

Cable positions + velocities: YES, from simulation. 84 policy inputs. Residual memory hidden; real sensing not demonstrated.

New hit gate: tip first, 5 cm radius, directed speed ≥ 4 m/s, angle ≤ 45 degrees.

Angle reward off. Retained D50 results use the older 30-degree gate.

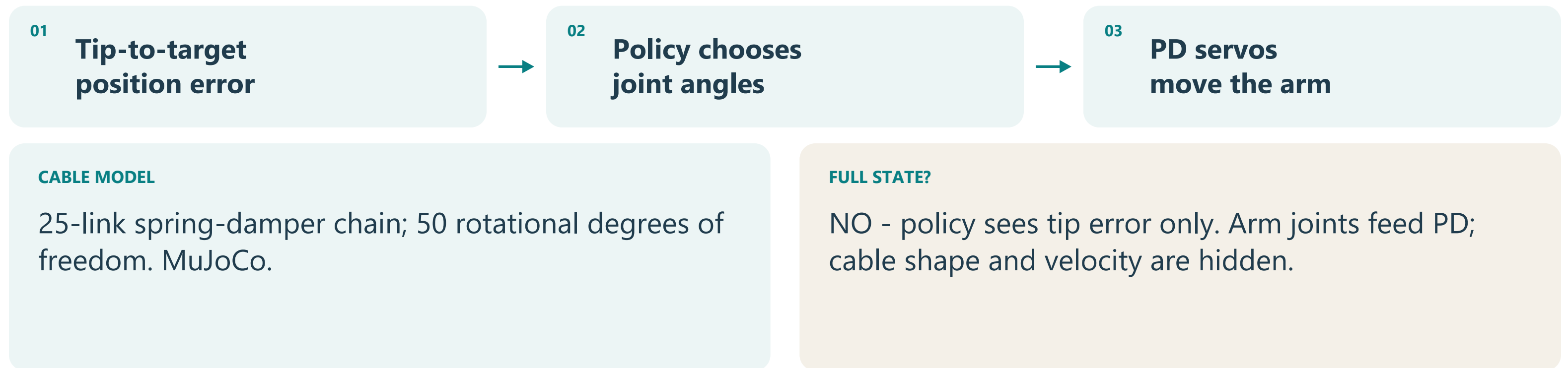
Learning a whip strike

Deep Reinforcement Learning of Robotic Manipulation for Whip Targeting

Goal: Teach a robot arm to hit a target with a whip.

Method: PPO learns from trial rewards; PD servos turn requested joint angles into torques.

HOW IT WORKS



Takeaway: The simulator knows the cable; the policy does not.

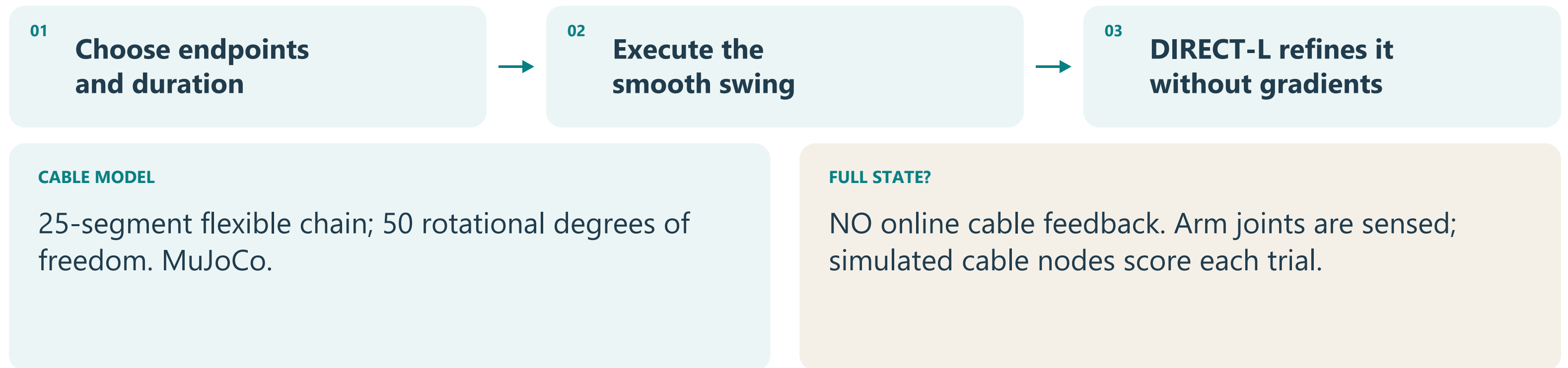
One smooth movement, nine parameters

Learning to manipulate a whip with simple primitive actions - A simulation study

Goal: Hit whip targets by optimizing a smooth arm swing.

Method: A motion primitive defines the swing. Impedance acts like a spring and damper around it.

HOW IT WORKS



Takeaway: A small search space simplifies control, while the simulator still models the whole cable.

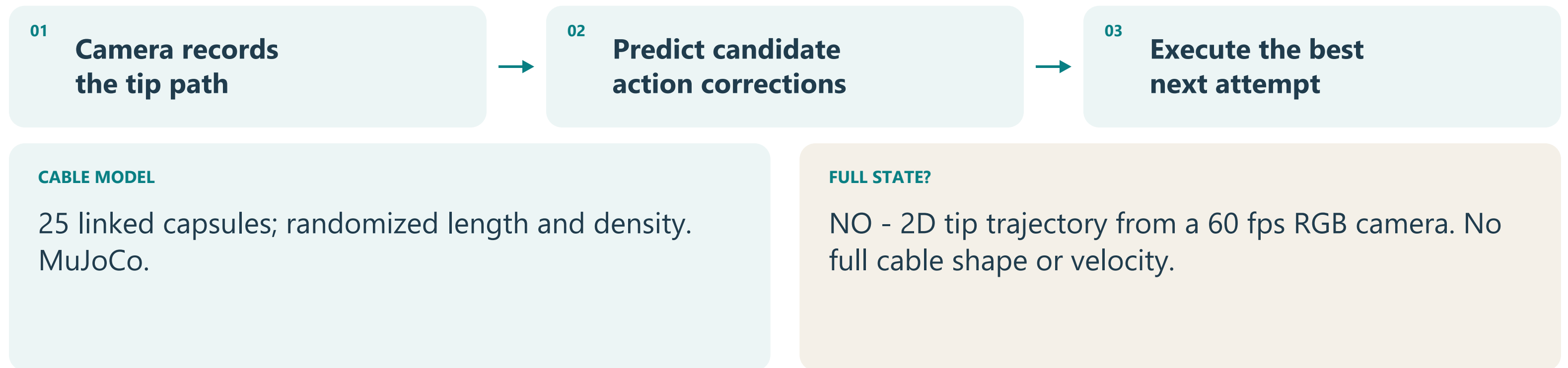
Improve the next attempt

Iterative Residual Policy: for Goal-Conditioned Dynamic Manipulation of Deformable Objects

Goal: Adapt a rope strike to different ropes using repeated trials.

Method: IRP learns how a small action correction changes the observed tip trajectory.

HOW IT WORKS



Takeaway: Corrections happen between attempts, not during a strike.

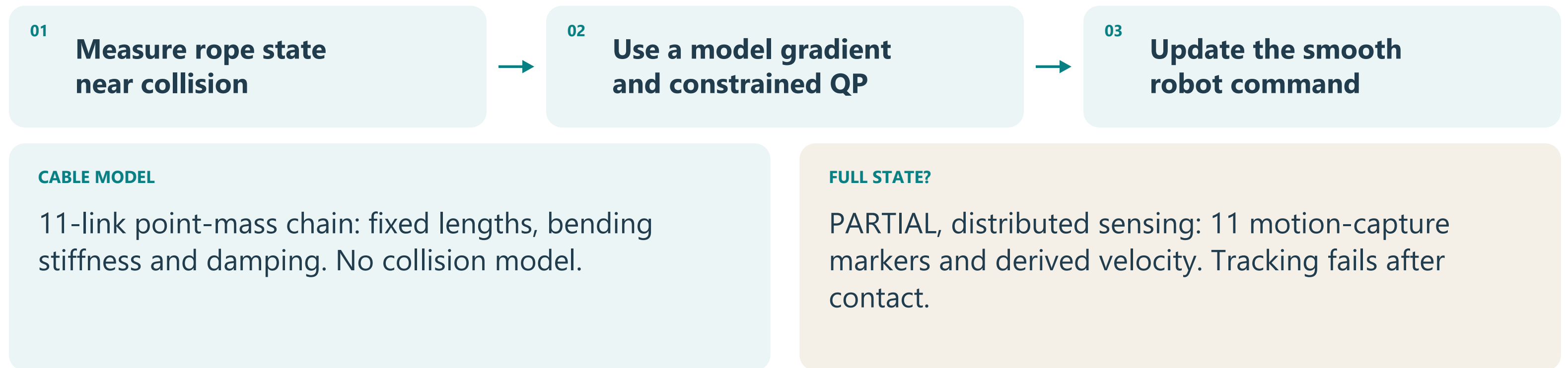
Learn the right state at the right event

Learning Dynamic Rope Manipulation Using Task-Level Iterative Learning Control

Goal: Learn a flying knot from one human demonstration.

Method: ILC converts measured event error into a command correction for the next trial.

HOW IT WORKS



Takeaway: Measure the rope across its length; learn from errors at the critical event.

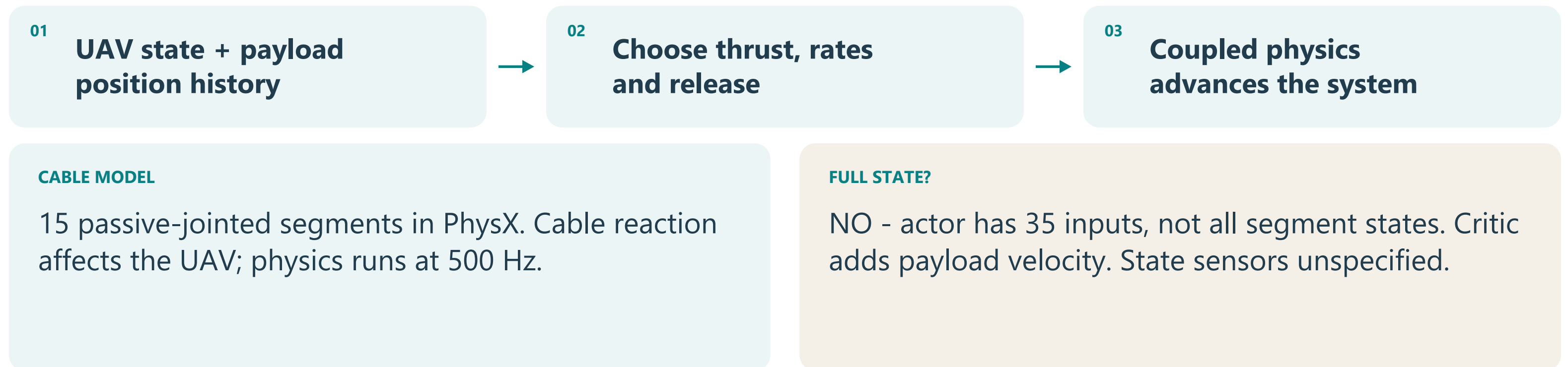
Throwing with a flying base

Learning to Throw: Agile and Accurate Cable-Suspended Payload Delivery with a Quadrotor

Goal: Swing and release a suspended payload toward a landing target.

Method: Asymmetric PPO gives the training critic more state information than the deployed actor.

HOW IT WORKS



Takeaway: Detailed coupled physics can train a controller with partial observations.

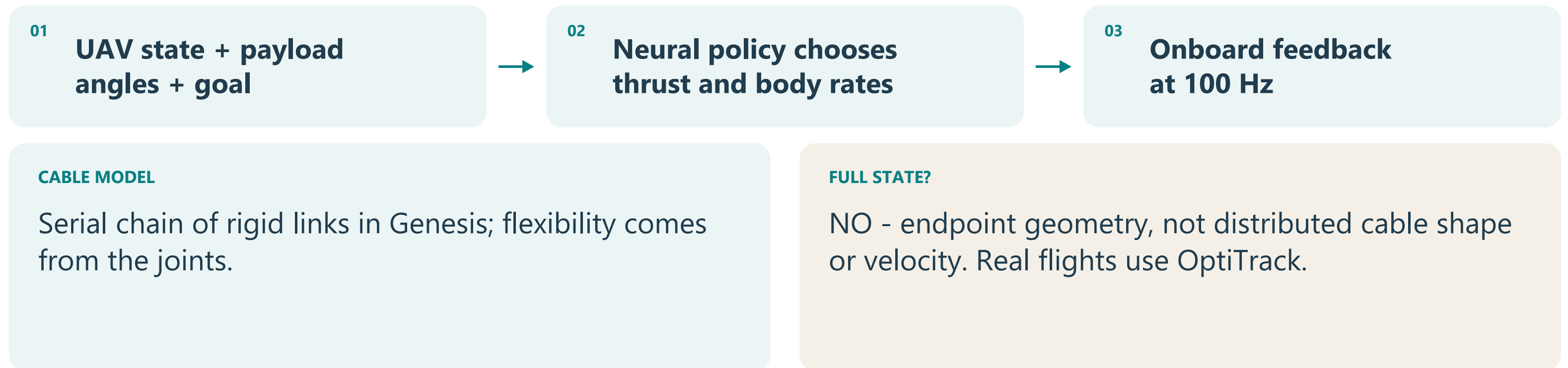
Agile flight from endpoint observations

FLARE: Agile Flights for Quadrotor Cable-Suspended Payload System via Reinforcement Learning

Goal: Move a suspended payload through targets and gates.

Method: PPO learns a direct observation-to-command mapping from simulated rewards.

HOW IT WORKS



Takeaway: Endpoint swing information may suffice for payload tasks; whip waves are a different challenge.

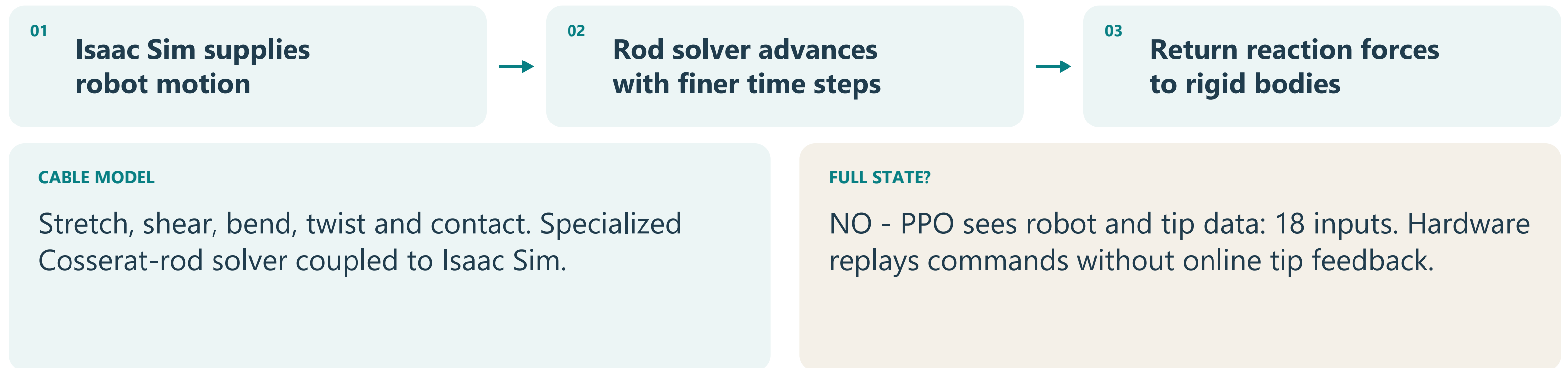
Better rod physics, transferable motion

DeformX: A Versatile Co-Simulation Framework for Deformable Linear Objects

Goal: Transfer learned rope-swinging commands to a real UR5e.

Method: Cosserat rods track centerline and cross-section orientation; deformation creates forces and moments.

HOW IT WORKS



Takeaway: Full rod simulation; partial policy observations; open-loop real execution.

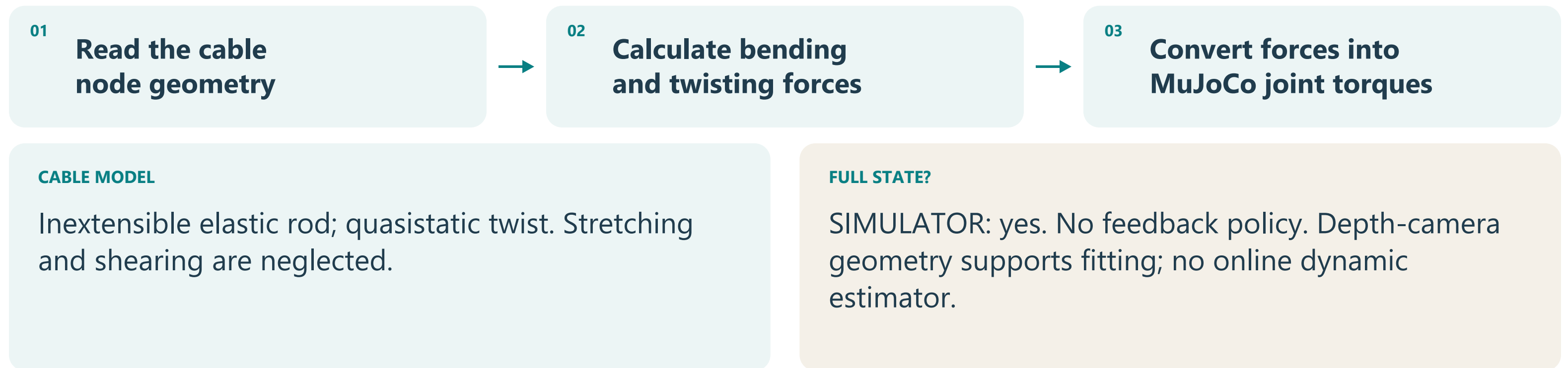
Identify the cable mechanics

Accurate Simulation and Parameter Identification of Deformable Linear Objects using Discrete Elastic Rods in Generalized Coordinates

Goal: Fit and simulate bending and twisting stiffness.

Method: DER computes forces from elastic-energy gradients: deformation stores energy, which drives restoring forces.

HOW IT WORKS



Takeaway: A mechanics and identification method; full-state control is not demonstrated.

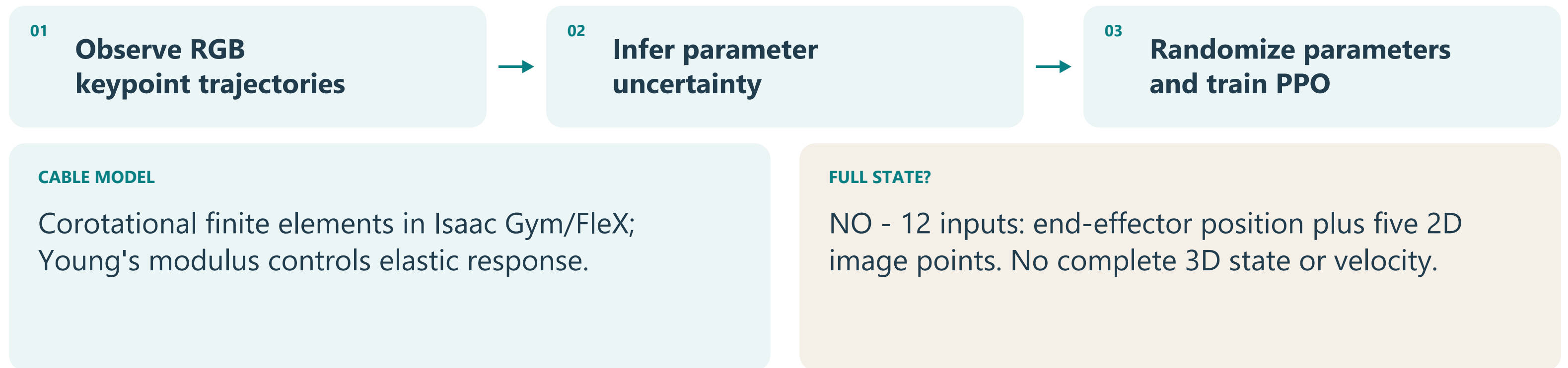
Learn physical uncertainty from vision

A Distributional Treatment of Real2Sim2Real for Object-Centric Agent Adaptation in Vision-Driven Deformable Linear Object Manipulation

Goal: Adapt deformable-object reaching to different stiffness and length.

Method: BayesSim infers plausible parameter distributions; PPO trains across samples from those distributions.

HOW IT WORKS



Takeaway: Estimating material parameters does not reveal the cable's instantaneous state.

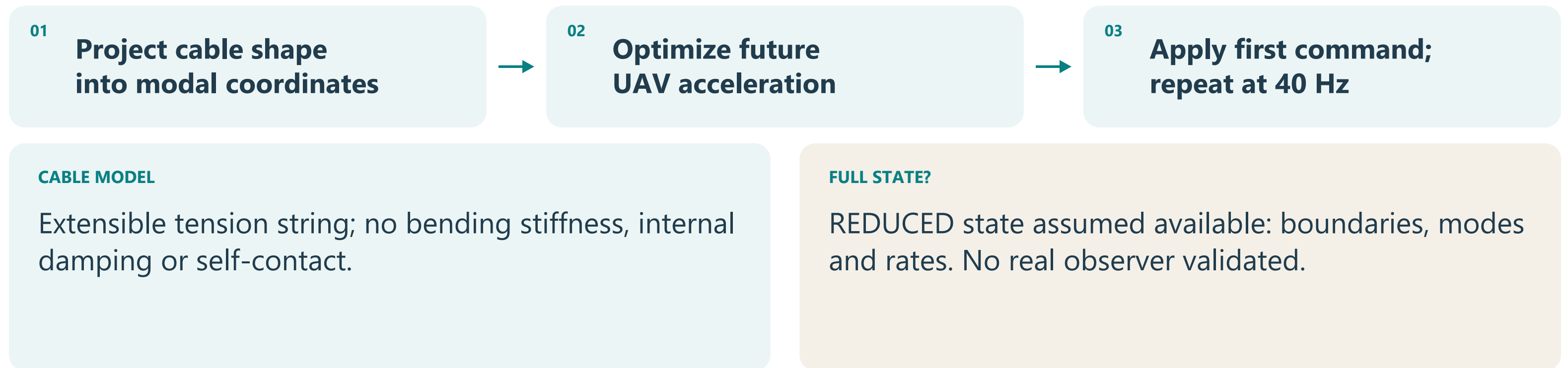
Predict with fewer cable variables

Nonlinear Predictive Control of the Continuum and Hybrid Dynamics of a Suspended Deformable Cable for Aerial Pick and Place

Goal: Track a cable tip through payload pickup and release.

Method: POD compresses deformation into modes. MPC predicts motion, optimizes commands, then replans.

HOW IT WORKS



Takeaway: A reduced controller state still needs a practical sensing solution.